

## **Reflection on Models Used:**

When working with Claude, we found that the model struggled with coming up with new use cases while generating multiple cases at once. This led to multiple cases output by the model being the same. To get around this limitation, we first generated a list of use case titles which would then be passed into the model one at a time, limiting the model to creating one use case per output. This had the benefit of working within the smaller context window of Claude, allowing it to focus on the use case at hand rather than generating multiple use cases at a time and hoping the output is unique and complete. Using this new method, we found that it did well sticking to the format of the example input given to the model, only making a minor mistake confusing UC1 as UC11.

While working with GPT-4o, we found that it generally adhered to the overall expected format of the Use Cases but it struggled with the finer details, such as labeling Alternative Flows and correctly numbering steps. It also had a habit of misclassifying assumptions as preconditions, indicating that there is a disconnect between the logical structure and the output. While the larger context window allowed for more extensive queries and outputs, it occasionally overcorrected when told not to assume something, often adding unnecessary preconditions such as the user having the app installed. This over specification was also seen in the way the model would tend to stick to using “the food delivery app” rather than using a more general identifier such as “the app.”

## **Pricing breakdown:**

- **Claude Sonnet 4: \$0.47**
- **OpenAI GPT-4o: \$0.43**